

VALIDATION PLAN

OPERATIONAL QUALIFICATION

RELIABILITY DEMONSTRATION TESTING MODULE

JR Validated Environment — Reliability Demonstration Testing Module v2.0

Field	Value
Document Number	JR-VP-RDT-001
Title	Validation Plan — Operational Qualification, Reliability Demonstration Testing Module
System	JR Validated Environment — RDT Module
Module Version	2.0
Document Version	3.0
Status	Approved
Effective Date	2026-10-01
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-04-22	Joep Rous	Initial release. Covers both RDT module scripts (jrc_rdt_plan, jrc_rdt_verify) in module v1.0. 25 test cases; all numeric assertions computed independently of R using closed-form Python arithmetic.
2.0	2026-05-18	Joep Rous	Added --report PV report test cases for jrc_rdt_verify (TC-RDT-VER-013, 014, 015) and corresponding UR-RDT-014. Total: 28 test cases, 14 user requirements.
3.0	2026-10-01	Joep Rous	jrscripsts v5.0.0, code review 2026-10: Bogey plans use the exact binomial criterion (TC-RDT-PLAN-006 updated); UR-RDT-010 restated (Weibayes counts all observed failures; actual test time and accel_factor); added TC-RDT-VER-016 (plan/verify round trip), TC-RDT-VER-017 (JSON escaping) and TC-RDT-VER-018 (late failures). Total: 31 test cases, 14 user requirements.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-05-18
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document defines the Validation Plan for the Operational Qualification (OQ) of the Reliability Demonstration Testing (RDT) module included in version 1.0 of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability between requirements and tests needed to demonstrate that each RDT script performs correctly for its intended use in FDA design verification under 21 CFR 820.30(f) and ISO 13485:2016 §7.3.6.

Execution of the tests defined in this plan, together with satisfactory results and documented evidence, constitutes the OQ for the RDT module.

2. Scope

2.1 In Scope

- Both R scripts in repos/rdt/R/ (jrc_rdt_plan.R, jrc_rdt_verify.R)
- Correct computation of required sample size (n) using the Bogey/binomial method (no Weibull shape assumption) and the Weibayes method (Weibull shape β provided)
- Correct effect of acceleration factor (AF) on sample size: n reduces by AF^β in Weibayes mode; Bogey mode is AF-independent
- Correct beta sensitivity table when accel_factor > 1
- Correct Binomial (Clopper-Pearson exact) verification: $R_{lower} = 1 - qbeta(C, k+1, n-k)$
- Correct Weibayes verification: $R_{demo} = \exp(-(TL/\eta_{demo})^\beta)$, where η_{demo} uses $\sum(t_{eff_i}^\beta)$
- Correct overall verdict logic: PASS if Binomial OR Weibayes passes
- Correct rejection of invalid inputs with informative error messages and non-zero exit
- Correct file I/O (CSV input, PNG output to ~/Downloads/)
- Verification that all RDT scripts enforce execution within the jrrun validated environment by checking RENV_PATHS_ROOT at startup

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by IQ; see JR-VP-001)
- Performance qualification (PQ) — conducted separately under specific test protocols
- User acceptance testing (UAT)
- Validation of R language interpreter and third-party packages (ggplot2) — package integrity is confirmed at installation via SHA256 (JR-IQ-001)
- Statistical validation of the underlying methods (Clopper-Pearson, Weibayes chi-squared pivot) — method references are cited in Section 3
- Weibull shape parameter estimation (β) — covered by jrc_weibull (separate script)

3. Regulatory Basis and References

3.1 Regulations

Reference	Title
21 CFR Part 820, §820.30(f)	Design Verification
21 CFR Part 820, §820.70(i)	Quality System Regulation — Automated Data Processing
21 CFR Part 11	Electronic Records; Electronic Signatures

ISO 13485:2016, §7.3.6	Design and development verification
------------------------	-------------------------------------

3.2 FDA Guidance

Reference	Title	Date
FDA GPSV	General Principles of Software Validation; Final Guidance for Industry and FDA Staff	January 2002
FDA CSA	Computer Software Assurance for Production and Quality System Software; Draft Guidance	September 2022

3.3 Statistical Methods

Reference	Title
Meeker, Hahn & Escobar (2017)	Statistical Intervals: A Guide for Practitioners and Researchers, 2nd ed. Wiley. Chapter 8 — Weibayes chi-squared pivot for reliability demonstration testing.
Nelson (2004)	Accelerated Testing: Statistical Models, Test Plans, and Data Analysis. Wiley. Weibayes method with known Weibull shape.
Clopper & Pearson (1934)	The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika 26(4), 404–413. Exact one-sided confidence bound on failure fraction: $F_{upper} = qbeta(C, k+1, n-k)$.

3.4 Internal Documents

Document Number	Title
JR-VP-001	Validation Plan — Installation Qualification, JR Validated Environment
JR-IQ-001	IQ Execution Evidence (docs/IQ_validation *.txt)
JR-VP-002	Validation Plan — OQ, Community Script Suite

4. Definitions and Abbreviations

Term	Definition
RDT	Reliability Demonstration Test. A statistically designed experiment to demonstrate that a device meets a pre-specified reliability claim (e.g. $R \geq 0.95$ at target life, 90% confidence).
R (reliability)	The proportion of units that survive to target life under use conditions. Distinct from R (gas constant) used in the Arrhenius equation.
C (confidence)	Statistical confidence level for the reliability claim (e.g. $0.90 = 90\%$).
k	Number of allowed (or observed) failures at or before target life.
n	Required (or actual) number of test units.
AF (accel_factor)	Life-extension multiplier. $t_{eff} = AF \times \text{target_life}$ per unit. $AF > 1$ means units are tested beyond target life, reducing n in Weibayes mode.
β (beta)	Weibull shape parameter. Governs the failure rate profile: $\beta = 1$ (random), $\beta > 1$ (wear-out), $\beta < 1$ (early failures). Required for Weibayes mode.
Bogey / Binomial	RDT method that makes no Weibull shape assumption. n is independent of AF. Exact binomial criterion: $n = \lceil qchisq(C, 2(k+1)) / (2 \times (-\ln R)) \rceil$. At $k=0$ this equals the zero-failure success-run formula.
Weibayes	RDT method using an assumed Weibull shape β . Incorporates AF: $n = \lceil qchisq(C, 2(k+1)) / (2 \times (-\ln R) \times AF^\beta) \rceil$.
Clopper-Pearson	Exact one-sided binomial confidence interval. $F_{upper} = qbeta(C, k+1, n-k)$; $R_{lower} = 1 - F_{upper}$. PASS if $R_{lower} \geq \text{reliability_claim}$.
T^*	Weibayes accumulated Weibull time: $T^* = \sum (t_{eff_i}^\beta)$ over all n units.
$T_{threshold}$	Minimum T^* required to demonstrate the claim: $T_{threshold} = qchisq(C, 2(k+1)) \times TL^\beta / (2 \times (-\ln R))$.
IQ	Installation Qualification
OQ	Operational Qualification
UR	User Requirement
TC	Test Case

jrrun	The validated bash entry-point script that sets the controlled environment before executing any script
PNG	Portable Network Graphics output file; plots saved to ~/Downloads/

5. User Requirements

The following user requirements define the expected behaviour of the RDT module. Each requirement is verifiable by the test cases in Section 7.

UR ID	Title	Requirement
UR-RDT-001	Bypass protection — RENV_PATHS_ROOT	All RDT scripts shall check the RENV_PATHS_ROOT environment variable at startup and call stop() with an informative message if it is not set, preventing execution outside the validated jrrun environment.
UR-RDT-002	Error handling — non-zero exit	All RDT scripts shall exit with a non-zero status code whenever an invalid input is detected (out-of-range parameter, missing required argument, bad CSV), and shall print an informative error message identifying the cause.
UR-RDT-003	Plan — Bogey/binomial sample size	jrc_rdt_plan shall compute required sample size for $k = 0$ to 5 allowed failures using the exact Bogey/binomial formula: $n = \lceil qchisq(C, 2(k+1)) / (2 \times (-\ln R)) \rceil$. At $k=0, C=0.90, R=0.95$ the result shall be $n = 45$.
UR-RDT-004	Plan — Weibayes sample size with acceleration factor	When --beta is provided, jrc_rdt_plan shall compute n using the Weibayes formula: $n = \lceil qchisq(C, 2(k+1)) / (2 \times (-\ln R) \times AF^\beta) \rceil$. At $AF=2, \beta=2, k=0, C=0.90, R=0.95$ the result shall be $n = 12$. At $AF=1$ the result is identical to Bogey regardless of β .
UR-RDT-005	Plan — beta sensitivity table	When --beta is provided and accel_factor > 1, jrc_rdt_plan shall display a beta sensitivity table showing n for a grid of beta values ($\beta \times [0.5, 0.75, 1.0, 1.25, 1.5]$, minimum 0.5) at $k=0$. At accel_factor = 1 the table shall note that n is beta-independent.
UR-RDT-006	Plan — k_allowed flag	jrc_rdt_plan shall accept --k_allowed K (non-negative integer, default 0). The row corresponding to $k = K$ shall be marked as the plan row in the output table.
UR-RDT-007	Plan — PNG output	jrc_rdt_plan shall save a two-panel PNG to ~/Downloads/ with a timestamp prefix and the suffix _jrc_rdt_plan.png. Panel 1: bar chart of n vs k . Panel 2: beta sensitivity line plot (when applicable).
UR-RDT-008	Verify — CSV input validation	jrc_rdt_verify shall accept a CSV file containing at least a time column and a status column (default names 'time' and 'status'; overridable via --time_col and --status_col). All time values must be positive numerics. Status values must be 0 (survived) or 1 (failed). Missing or wrong-named columns shall cause a non-zero exit.
UR-RDT-009	Verify — Binomial (Clopper-Pearson) verification	jrc_rdt_verify shall compute the exact Clopper-Pearson upper confidence bound: $F_{upper} = qbeta(C, k+1, n-k)$, where k = failures with $t_{eff} \leq target_life$. $R_{lower} = 1 - F_{upper}$. When $n-k = 0$ (all units failed), $F_{upper} = 1$ and $R_{lower} = 0$. PASS if $R_{lower} \geq reliability_claim$. At $n=45, k=0, C=0.90$ the result shall be $R_{lower} \approx 0.9501$.
UR-RDT-010	Verify — Weibayes verification	When --beta is provided, jrc_rdt_verify shall compute the Weibayes demonstrated reliability: $T^* = \Sigma(t_{eff_i}^\beta)$; $\eta_{demo} = (2T^* / qchisq(C, 2(r+1)))^{1/\beta}$ with r = ALL observed failures (including failures after target life); $R_{demo} = \exp(-(TL/\eta_{demo})^\beta)$. The time column holds actual test time and $t_{eff} = time \times accel_factor$ (stress acceleration). PASS if $R_{demo} \geq reliability_claim$. At $n=45, \beta=2, t=5000, k=0, C=0.90$ the result shall be $R_{demo} \approx 0.9501$.
UR-RDT-011	Verify — overall verdict	jrc_rdt_verify shall report an overall verdict of PASS if either the Binomial or the Weibayes method (when applicable) produces a PASS result. When the two methods disagree, a note shall be

		printed.
UR-RDT-012	Verify — exit code convention	jrc_rdt_verify shall exit 0 for both PASS and FAIL verdicts. Non-zero exit is reserved for input errors and runtime failures.
UR-RDT-013	Verify — PNG output	jrc_rdt_verify shall save a two-panel PNG to ~/Downloads/ with a timestamp prefix and the suffix _jrc_rdt_verify.png. Panel 1: unit timeline (survivors = +, failures = x). Panel 2: demonstrated reliability bars (Binomial and Weibayes if applicable).
UR-RDT-014	Verify — --report output	When invoked with the --report flag, jrc_rdt_verify shall write a verification report to ~/Downloads/. If jr_pack is installed the report is generated as a .docx file; otherwise the intermediate .html file remains. A JSON sidecar shall contain report_type == "rdt" and verdict_pass as a boolean reflecting the overall PASS/FAIL verdict.

6. Test Environment

Component	Specification
Operating System	macOS 15.x (Apple Silicon) or Windows 10/11 (Git Bash)
R Version	As installed per JR-IQ-001 (R 4.3.x or later)
Python Version	As installed per JR-IQ-001 (Python 3.11.x or later)
Test Framework	pytest 8.x
Test runner	repos/rdt/admin_rdt_oq
Test files	repos/rdt/oq/test_rdt_plan.py (13 tests), repos/rdt/oq/test_rdt_verify.py (12 tests)
Test data	repos/rdt/oq/data/ (5 CSV fixtures)
OQ venv	~/venvs/<PROJECT_ID>_oq/ (shared with other module OQ suites)
Output directory	~/Downloads/ (PNG files verified by mtime glob)
Entry point	bin/jrrun (integrity-checked execution)

7. Test Cases

All test cases are implemented in repos/rdt/oq/test_rdt_plan.py and repos/rdt/oq/test_rdt_verify.py. Test data files reside in repos/rdt/oq/data/. Numerical reference values are computed independently of the R scripts using pure Python arithmetic (math.log, math.exp, math.ceil, bisection on exact CDFs) — no R or scipy is used in the reference calculations.

7.1 jrc_rdt_plan

TC-RDT-PLAN-001

Tests: UR-RDT-003

Command: jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000

Rationale: Bogey mode (no --beta): verifies the mode header appears in output.

Pass criterion: exit 0; 'Binomial' in output

TC-RDT-PLAN-002

Tests: UR-RDT-004

Command: jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0

Rationale: Weibayes mode: verifies the mode header appears in output.

Pass criterion: exit 0; 'Weibayes' in output

TC-RDT-PLAN-003

Tests: UR-RDT-003

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: Known numeric, Bogey $k=0$. Reference: $n = \text{ceil}(\log(0.10)/\log(0.95)) = 45$. Computed in pure Python (`math.ceil`, `math.log`), independent of R.

Pass criterion: exit 0; 'n = 45' in output

TC-RDT-PLAN-004

Tests: UR-RDT-004

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0`

Rationale: Weibayes $k=0$ $AF=1$: $AF^\beta = 1^2 = 1$ for any β . Formula reduces to Bogey $\rightarrow n=45$.

Pass criterion: exit 0; 'n = 45' in output

TC-RDT-PLAN-005

Tests: UR-RDT-004

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0 --accel_factor 2.0`

Rationale: Weibayes $k=0$, $\beta=2$, $AF=2$. Reference: $n = \text{ceil}(\log(0.10)/(\log(0.95)*4)) = 12$. Computed in pure Python, independent of R.

Pass criterion: exit 0; 'n = 12' in output

TC-RDT-PLAN-006

Tests: UR-RDT-003, UR-RDT-006

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --k_allowed 1`

Rationale: $k=1$ Bogey, exact binomial (same Clopper-Pearson criterion as `jrc_rdt_verify`). Reference in pure Python: smallest n with $P(X \leq 1 \mid n, 0.05) \leq 0.10 \rightarrow n = 77$ ($n = 76$ gives 0.1014). Plan marker '<- plan' on $k=1$ row.

Pass criterion: exit 0; 'n = 77' in output; '<- plan' in output

TC-RDT-PLAN-007

Tests: UR-RDT-002

Command: `jrrun jrc_rdt_plan.R --help`

Rationale: `--help` flag: exits 0 with usage text.

Pass criterion: exit 0; 'Usage' in output

TC-RDT-PLAN-008

Tests: UR-RDT-005

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0 --accel_factor 2.0`

Rationale: Beta sensitivity table: requires `--beta` and `accel_factor > 1`.

Pass criterion: exit 0; 'Beta Sensitivity' in output

TC-RDT-PLAN-009

Tests: UR-RDT-007

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: PNG output to `~/Downloads/`.

Pass criterion: exit 0; `*_jrc_rdt_plan.png` found in `~/Downloads/` after run

TC-RDT-PLAN-010

Tests: UR-RDT-002

Command: `jrrun jrc_rdt_plan.R --reliability 1.5 --confidence 0.90 --target_life 5000`

Rationale: `--reliability = 1.5` (outside (0,1)).

Pass criterion: exit non-zero

TC-RDT-PLAN-011

Tests: UR-RDT-002

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0 --target_life 5000`

Rationale: `--confidence = 0` (outside (0,1)).

Pass criterion: exit non-zero

TC-RDT-PLAN-012

Tests: UR-RDT-002

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --target_life 5000`

Rationale: Missing required argument `--confidence`.

Pass criterion: exit non-zero

TC-RDT-PLAN-013

Tests: UR-RDT-001

Command: `Rscript repos/rdt/R/jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 (RENV_PATHS_ROOT unset)`

Rationale: Direct Rscript call bypasses jrrun.

Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

7.2 jrc_rdt_verify

TC-RDT-VER-001

Tests: UR-RDT-008, UR-RDT-009, UR-RDT-012

Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: `rdt_verify_pass.csv`: 45 units, all survived (`status=0`, `t=5000`). Happy path.

Pass criterion: exit 0; 'Verdict' in output

TC-RDT-VER-002

Tests: UR-RDT-009, UR-RDT-012

Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: Numeric assertion: $n=45$, $k=0$. $R_lower = 1 - (1-0.90)^{(1/45)} \approx 0.9501$. Reference computed in pure Python (exact Beta(1,n) ppf). Tolerance ± 0.0002 .

Pass criterion: exit 0; 'PASS' in output; $R_lower = 0.9501 \pm 0.0002$

TC-RDT-VER-003

Tests: UR-RDT-009, UR-RDT-012

Command: `jrrun jrc_rdt_verify.R rdt_verify_fail.csv --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: `rdt_verify_fail.csv`: 20 units all survived. $n=20$, $k=0$. $R_lower = 1 - 0.10^{(1/20)} \approx 0.8913 < 0.95$. FAIL verdict; script exits 0 (FAIL is not a script error).

Pass criterion: exit 0; 'FAIL' in output; $R_lower = 0.8913 \pm 0.0002$

TC-RDT-VER-004

Tests: UR-RDT-010, UR-RDT-011

Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0`

Rationale: Weibayes with $n=45$, $\beta=2$, $t=5000$, $k=0$. Reference: $T^* = 45 \times 5000^2 = 1.125 \times 10^9$; $qchisq(0.90, 2) = -2 \times \ln(0.10) \approx 4.6052$ (exact). $\eta_demo = (2 \times T^* / 4.6052)^{0.5} \approx 22104$; $R_demo = \exp(-(5000/22104)^2) \approx 0.9501$. Tolerance ± 0.0002 .

Pass criterion: exit 0; 'Weibayes' in output; 'PASS' in output; $R_demo = 0.9501 \pm 0.0002$

TC-RDT-VER-005

Tests: UR-RDT-013

Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: PNG output to ~/Downloads/.

Pass criterion: exit 0; *_jrc_rdt_verify.png found in ~/Downloads/ after run

TC-RDT-VER-006

Tests: UR-RDT-009

Command: `jrrun jrc_rdt_verify.R rdt_verify_all_failed_early.csv --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: rdt_verify_all_failed_early.csv: 10 units, all failed at $t=2500 < \text{target_life}=5000$. Edge case: $k=n=10$, $n-k=0$. Script sets $F_{\text{upper}}=1.0$, $R_{\text{lower}}=0.0$. Verified by extract on 'R_lower=' label.

Pass criterion: exit 0; 'FAIL' in output; $R_{\text{lower}} < 0.01$

TC-RDT-VER-007

Tests: UR-RDT-009

Command: `jrrun jrc_rdt_verify.R rdt_verify_zero_failures.csv --reliability 0.90 --confidence 0.90 --target_life 5000`

Rationale: rdt_verify_zero_failures.csv: 50 units, all survived. $R_{\text{lower}} = 1 - 0.10^{(1/50)} \approx 0.9550$. Reference computed in pure Python. Assert $R_{\text{lower}} \in [0.953, 0.959]$.

Pass criterion: exit 0; R_{lower} in $[0.953, 0.959]$

TC-RDT-VER-008

Tests: UR-RDT-002

Command: `jrrun jrc_rdt_verify.R` (no arguments)

Rationale: --help shown when no arguments provided; exits 0.

Pass criterion: exit 0; 'Usage' in output

TC-RDT-VER-009

Tests: UR-RDT-008, UR-RDT-002

Command: `jrrun jrc_rdt_verify.R /tmp/no_such_file_rdt_oq.csv --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: Non-existent file path.

Pass criterion: exit non-zero; 'not found' or 'File' in output

TC-RDT-VER-010

Tests: UR-RDT-008

Command: `jrrun jrc_rdt_verify.R rdt_verify_missing_col.csv --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: rdt_verify_missing_col.csv has column 'hours' not 'time'. Error message must name the missing column.

Pass criterion: exit non-zero; 'time' in error output

TC-RDT-VER-011

Tests: UR-RDT-008

Command: `jrrun jrc_rdt_verify.R <inline temp CSV with non-numeric time> --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: Temp CSV created in-test with `time='abc'`. Non-numeric time value.

Pass criterion: exit non-zero

TC-RDT-VER-012

Tests: UR-RDT-001

Command: `Rscript repos/rdt/R/jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000 (RENV_PATHS_ROOT unset)`

Rationale: Direct Rscript call bypasses jrrun.

Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

TC-RDT-VER-013

Tests: UR-RDT-014

Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000 --report`

Rationale: --report flag triggers verification report generation. Accepts .docx (jr_pack full install) or .html (template-only install).

Pass criterion: exit 0; file matching *_rdt_verification_report.docx or *_rdt_verification_report.html found in ~/Downloads/ with recent mtime.

TC-RDT-VER-014

Tests: UR-RDT-014

Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000 --report`

Rationale: When jr_pack is fully installed the JSON sidecar is consumed after .docx generation.

Pass criterion: exit 0; file matching *_rdt_verification_report.docx or *_rdt_verification_report_data.json found in ~/Downloads/ with recent mtime.

TC-RDT-VER-015

Tests: UR-RDT-014

Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000 --report`

Rationale: JSON sidecar must carry report_type == "rdt" and verdict_pass == True. TC-RDT-VER-002 confirms PASS for rdt_verify_pass.csv.

Pass criterion: exit 0; .docx or JSON found; if JSON present: report_type == "rdt" and verdict_pass is True.

TC-RDT-VER-016

Tests: UR-RDT-003, UR-RDT-009

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --k_allowed 1; then jrc_rdt_verify.R on generated CSVs (n and n - 1 units, 1 failure at t = 2500)`

Rationale: Plan → verify round trip (code review 2026-10, RDT-01): plan and verify apply the same exact binomial criterion, so the planned n (read from the plan output) passes with 1 failure and n - 1 fails.

Pass criterion: Plan exit 0 with a k = 1 plan row; verify on n units: 'Overall Verdict: PASS'; verify on n - 1 units: 'Overall Verdict: FAIL'.

TC-RDT-VER-017

Tests: UR-RDT-014

Command: `jrrun jrc_rdt_verify.R 'rdt "lot 7".csv' --reliability 0.95 --confidence 0.90 --target_life 5000 --report`

Rationale: Input file name contains double quotes (code review 2026-10, X-04): the report JSON must escape them. Skipped when the Validation Pack is not installed.

Pass criterion: exit 0; .docx or JSON found; if JSON present it parses and input_file == 'rdt "lot 7".csv'.

TC-RDT-VER-018

Tests: UR-RDT-010

Command: `jrrun jrc_rdt_verify.R <temp CSV: 40 units at 6000, 2 failed> --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2`

Rationale: Late failures count in the Weibayes df (code review 2026-10, RDT-04): $r = 2$, $T^* = 1.44e9$, $\chi^2(0.90, 6) = 10.6446$, $\eta = 16449$, $R_{\text{demo}} = 0.9117$ ($r = 0$ gave 0.9608, PASS).

Pass criterion: exit 0; 'Failures (all, r): 2'; $R_{\text{demo}} = 0.9117 \pm 0.0001$; 'Overall Verdict: FAIL'.

8. Requirements Traceability Matrix

Each user requirement is covered by at least one test case.

UR ID	Title	Test Cases
UR-RDT-001	Bypass protection (RENV_PATHS_ROOT)	TC-RDT-PLAN-013, TC-RDT-VER-012
UR-RDT-002	Non-zero exit on error	TC-RDT-PLAN-007, TC-RDT-PLAN-010, TC-RDT-PLAN-011, TC-RDT-PLAN-012, TC-RDT-VER-008, TC-RDT-VER-009, TC-RDT-VER-010, TC-RDT-VER-011
UR-RDT-003	Plan — Bogey/binomial sample size	TC-RDT-PLAN-001, TC-RDT-PLAN-003, TC-RDT-PLAN-006, TC-RDT-VER-016
UR-RDT-004	Plan — Weibayes sample size with AF	TC-RDT-PLAN-002, TC-RDT-PLAN-004, TC-RDT-PLAN-005
UR-RDT-005	Plan — beta sensitivity table	TC-RDT-PLAN-008
UR-RDT-006	Plan — k_allowed flag	TC-RDT-PLAN-006
UR-RDT-007	Plan — PNG output	TC-RDT-PLAN-009
UR-RDT-008	Verify — CSV input validation	TC-RDT-VER-001, TC-RDT-VER-009, TC-RDT-VER-010, TC-RDT-VER-011
UR-RDT-009	Verify — Binomial (Clopper-Pearson)	TC-RDT-VER-001, TC-RDT-VER-002, TC-RDT-VER-003, TC-RDT-VER-006, TC-RDT-VER-007, TC-RDT-VER-016
UR-RDT-010	Verify — Weibayes verification	TC-RDT-VER-004, TC-RDT-VER-018
UR-RDT-011	Verify — overall verdict	TC-RDT-VER-004
UR-RDT-012	Verify — exit code convention	TC-RDT-VER-001, TC-RDT-VER-002, TC-RDT-VER-003, TC-RDT-VER-008, TC-RDT-VER-009
UR-RDT-013	Verify — PNG output	TC-RDT-VER-005
UR-RDT-014	Verify — --report output	TC-RDT-VER-013, TC-RDT-VER-014, TC-RDT-VER-015, TC-RDT-VER-017

9. Acceptance Criteria

The RDT module OQ is considered PASS when all of the following conditions are satisfied:

- All 31 automated test cases execute with the result PASSED.
- Zero test cases result in FAILED or ERROR.
- The OQ evidence file is generated and preserved at
~/jrscrip</PROJECT_ID>/validation/rdt_oq_execution_<timestamp>.txt.
- All test cases are executed using the validated jrrun entry-point (bypass protection verified for both scripts).
- No open deviations remain unresolved at the time of approval.

Any test failure shall be investigated, documented as a deviation, resolved, and the affected test(s) re-executed. The OQ may not be closed until all deviations are resolved.